Mikkel Paltorp

mikkel.paltorp@gmail.com | github.com/mipals | mipals.github.io

Work Experience

PostDoctoral Researcher (SciML for Structural Health Monitoring) 2025 — Present
Technical University of Denmark (DTU Compute) *Kgs. Lyngby, Denmark*

- Applying transfer learning techniques to predict the structural health of offshore structures.
- Keywords: Kalman Filtering, Time Series Classification, Gaussian Processes.

PostDoctoral Researcher (Conic Optimization for Nonlinear Finite Element Analysis) 2024
Technical University of Denmark (DTU Compute) / OPTUM CE *Kgs. Lyngby / Copenhagen, Denmark*

- Developed a custom inhouse conic interior point solver. Worked with sparse linear algebra, Rust, C++, and Julia.
- Collaboration with industry partner (OPTUM CE) implementing our solution into their FELA software.

Ph.D in Computational Acoustics 2020 — 2024
Technical University of Denmark (DTU Electro) *Kgs. Lyngby, Denmark*

- Lead the research into applying Fast Multipole Methods and H-matrices to viscothermal acoustics. Using these methods we were able to solve problems orders of magnitudes larger than previously possible.
- Developed a Boundary Element code in Julia. The code is now used at the Technical University of Munich.
- Taught Finite & Boundary Element Methods and guided student projects in the course Numerical Acoustics.

Education

Technical University of Denmark (DTU) Kongens Lyngby, Denmark
B.Sc.Eng & M.Sc.Eng in Computational Science and Engineering *2014 — 2020*

- Took an interest in differential equations, numerical methods, and optimization algorithms.
- For my thesis I implemented scalable (i.e. linear time and storage) algorithms for Gaussian processes.

McGill University Montreal, Canada
Exchange Semester *2016 Fall Semester*

- Took courses in complex analysis, non-linear and partial differential equations, and mathematical biology.

Projects

BoundaryIntegralEquations.jl / FMM2D.jl, Creator

- Utilizes the Flatiron Institute Fast Multipole Libraries. Developed a wrapper for the 2D library (FMM2D.jl).
- Includes SOTA techniques for large-scale viscothermal acoustics that mixes sparse linear algebra with the FMM.
- Includes automatically documentation generation using Documenter.jl and Literate.jl.

Pardiso.jl, Maintainer

- A wrapper for the sparse direct solvers MKL/Panua-Pardiso. The solvers are used throughout the Julia ecosystem.

Skills

- **Programming Languages**: Julia, Python, C, R, Rust, MatLab
- **Technologies**: PyTorch, JAX, Git, LaTeX, Typst, CI/CD, uv, Mathematical Programming (JuMP / CVXPY)

Conferences and Workshops

International Congress on Acoustics (ICA2022) Gyeongju, South Korea
A reduced order model including viscothermal losses *Oral Presentation*

KAIST-DTU Workshop 2022 Daejeon, South Korea
Towards Large-scale Viscothermal Acoustics Simulations using the Boundary Element Method *Oral Presentation*

Forum Acusticum 2023 Turin, Italy
An Open-Source Boundary Element Framework for Large-scale viscothermal acoustics *Oral Presentation*